

Day 19 - Lecture 19.15: Relational & Matrix Plots - Heatmaps

1. Overview & Purpose

A **Heatmap** represents 2D tabular or matrix data where individual values are encoded as **color variations**. Primary use cases in Machine Learning and Data Science:

1. **Correlation Matrix:** Detecting multi-collinearity and linear relationships between feature columns.
 2. **Pivot Tables & Cross-Tabulations:** Analyzing interactions between two categorical dimensions (e.g., Passengers across Years and Months).
 3. **Confusion Matrices:** Evaluating classification model performance.
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2. Pivot Tables & Matrix Transformation

Raw time-series data often comes in "long form":

year	month	passengers
1949	Jan	112
1949	Feb	118

To plot a heatmap, we must reshape it into "wide/matrix form" using Pandas `.pivot()` :

```
flights_pivot = flights.pivot(index="month", columns="year", values="passengers")
```

```
1949  1950  ...  1960 Jan   112   115  ...  417 Feb   118   126  ...  391
```

3. Key Parameters of `sns.heatmap()`

```
sns.heatmap(  
    data,                # 2D rectangular dataset (DataFrame or 2D array)  
    annot=True,          # If True, prints numerical values inside each cell  
    fmt="d",             # Format string: "d" for integer, ".2f" for 2 decimals  
    cmap="coolwarm",     # Colormap ('viridis', 'coolwarm', 'YlGnBu', 'magma')  
    linewidths=0.5,     # Width of lines separating each cell  
    cbar=True            # Show colorbar legend  
)
```

4. Code Implementation

```
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Pivot Table Heatmap: Flight Passengers
flights = sns.load_dataset("flights")
flights_pivot = flights.pivot(index="month", columns="year", values="passengers")

plt.figure(figsize=(11, 7))
sns.heatmap(
    flights_pivot,
    cmap="YlGnBu",
    annot=True,
    fmt="d",
    linewidths=0.5
)
plt.title("Monthly Airline Passengers from 1949 to 1960", fontsize=15, fontweight="bold")
plt.xlabel("Year", fontsize=12)
plt.ylabel("Month", fontsize=12)
plt.tight_layout()
plt.show()

# 2. Correlation Matrix Heatmap: Tips Dataset
tips = sns.load_dataset("tips")
# Select only numeric columns for correlation
numeric_tips = tips.select_dtypes(include=['float64', 'int64'])
corr_matrix = numeric_tips.corr()

plt.figure(figsize=(7, 5))
sns.heatmap(
    corr_matrix,
    cmap="coolwarm",
    annot=True,
    fmt=".2f",
    vmin=-1, vmax=1,
    center=0,
    square=True
)
plt.title("Correlation Matrix (Tips Dataset)", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.show()
```

5. Key Takeaways

- Diverging vs. Sequential Colormaps:

- For **Correlations** (ranging from -1 to $+1$), use a **diverging colormap** like `coolwarm` with `center=0`.
- For **Magnitudes / Counts** (ranging from 0 to large values), use a **sequential colormap** like `YlGnBu` or `viridis`.
- Always set `fmt="d"` for integers and `fmt=".2f"` for floats; otherwise, scientific notation will clutter cells.